MATH 115 Sample Midterm 1

Questions 1 - 6 are multiple choice questions. For each of these questions, there is exactly one correct answer.

1. Let $\vec{x}, \vec{y} \in \mathbb{R}^2$. Which of the following statements is true?

   (a) If $\vec{x} \cdot \vec{y} = 0$, then $\vec{x} = \vec{0}$ or $\vec{y} = \vec{0}$.

   (b) If $\vec{x} \cdot \vec{z} = \vec{y} \cdot \vec{z}$ for some $\vec{z} \in \mathbb{R}^2$, then $\vec{x} = \vec{y}$.

   (c) If $\|\vec{x}\| = \|\vec{y}\|$, then $\vec{x} = \vec{y}$.

   (d) If $\vec{x} \cdot \vec{z} > 0$ and $\vec{z} \cdot \vec{y} > 0$ for some $\vec{z} \in \mathbb{R}^2$, then $\vec{x} \cdot \vec{y} > 0$.

   (e) None of the above.

2. Let P_1 and P_2 be points in $\mathbb{R}^3$ and $\vec{d}_1, \vec{d}_2 \in \mathbb{R}^3$ be nonzero. Consider the lines

$$L_1 \text{ with vector equation } \vec{x} = \overrightarrow{OP_1} + t\vec{d}_1, \quad t \in \mathbb{R}$$
$$L_2 \text{ with vector equation } \vec{x} = \overrightarrow{OP_2} + s\vec{d}_2, \quad s \in \mathbb{R}.$$

   Which of the following must be true?

   (a) If L_1 passes through the origin, then $\overrightarrow{OP_1} = \vec{0}$.

   (b) There is a plane in $\mathbb{R}^3$ that contains L_1 and L_2.

   (c) If L_1 and L_2 are parallel and A is any point on L_1 and B is any point on L_2, then the shortest distance between L_1 and L_2 is given by $\| \operatorname{perp}_{\vec{d}_1} \overrightarrow{AB} \|$.

   (d) The equation $\vec{x} = \overrightarrow{OP} + t\vec{d}_1 + s\vec{d}_2$ with $s, t \in \mathbb{R}$ is the vector equation of a plane through the point P.

   (e) None of the above statements are true.

3. Let $A \in M_{m \times n}(\mathbb{R})$ and let R be the reduced row echelon form of A. Which of the following statements must be true?

   (a) If R has a row of zeros, then A has a row of zeros.

   (b) If $\operatorname{rank}(A) < n$, then R has at least one column of zeros.

   (c) If $\operatorname{rank}(A) < m$, then R has at least one row of zeros.

   (d) R^T is the reduced row echelon form of A^T.

4. Let $A \in M_{m\times n}(\mathbb{R})$ and $\vec{b} \in \mathbb{R}^m$, and consider the system of equations given by $A\vec{x} = \vec{b}$. Which of the following must be true?

   (a) If $\vec{y}$ and $\vec{z}$ are solutions to $A\vec{x} = \vec{b}$, then so too is $\vec{y} + \vec{z}$.

   (b) If $A\vec{x} = \vec{b}$ is consistent, then $\text{rank}(A) = m$.

   (c) If $\vec{b}$ can be expressed as a linear combination of the rows of A, then $A\vec{x} = \vec{b}$ is consistent.

   (d) If $A\vec{x} = \vec{0}$ (that is, if $\vec{b} = \vec{0}_{\mathbb{R}^m}$), then either $A = 0_{m\times n}$ or $\vec{x} = \vec{0}_{\mathbb{R}^n}$.

   (e) None of the above.

5. Let $A \in M_{n\times m}(\mathbb{R})$, $B \in M_{p\times n}(\mathbb{R})$, $C \in M_{\ell\times k}(\mathbb{R})$ and $D \in M_{p\times k}(\mathbb{R})$. Which of the following is true?

   (a) $(CD^TBA)^T \in M_{\ell\times m}(\mathbb{R})$.

   (b) $(CD^TBA)^T \in M_{m\times p}(\mathbb{R})$.

   (c) $(CD^TBA)^T \in M_{m\times \ell}(\mathbb{R})$.

   (d) The product $(CD^TBA)^T$ is not defined.

   (e) None of the above.

6. Let $A \in M_{3\times 1}(\mathbb{R})$ and $B \in M_{1\times 3}(\mathbb{R})$ be two nonzero matrices. Then $\text{rank}(AB)$ equals

   (a) 0.

   (b) 1.

   (c) 2.

   (d) 3.

Questions 7 - 13 are long answer questions.

7. Consider the vectors
$$\vec{u} = \begin{bmatrix} 0 \\ -3 \\ -1 \end{bmatrix} \qquad \vec{v} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}.$$
   (a) Compute $\text{proj}_{\vec{u}}\, \vec{v}$ and $\text{perp}_{\vec{u}}\, \vec{v}$.
   (b) Compute $\vec{u} \times \vec{v}$.
   (c) Compute $\cos\theta$, where θ is the angle determined by $\vec{u}$ and $\vec{v}$.
   (d) Find a unit vector $\widehat{w}$ in the direction of $\vec{u} + \vec{v}$.

8. Find the shortest distance from the line with vector equation
$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -1 \end{bmatrix} + t \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix} \quad t \in \mathbb{R}$$
   to the plane with scalar equation $3x_1 + x_2 = 2$.

9. For $\ell \in \mathbb{R}$, consider a system of equations with augmented matrix
$$[\, A \,|\, \vec{b}\,] = \left[\begin{array}{ccc|c} 5 & 2 & -3 & 7 \\ 0 & 1 & 3 & 6 \\ 0 & 0 & \ell^2 - 4 & \ell + 2 \end{array}\right]$$
   With justification, find all $\ell \in \mathbb{R}$ so that the system has
   (a) one solution,
   (b) no solution,
   (c) infinitely many solutions.

10. Consider the two planes in $\mathbb{R}^3$ whose equations are
$$3x - 2y + 2z = 11 \qquad \text{and} \qquad 2x - 5y - 6z = 11.$$
    The intersection of these planes is a line in $\mathbb{R}^3$. Find a *vector* equation of this line.

11. Find the inverse of $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 2 & 5 \end{bmatrix}$.

12. (a) What does the vector equation

$$\vec{x} = r \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + s \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + t \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}, \quad r, s, t \in \mathbb{R}$$

represent geometrically?

(b) Can every vector $\vec{x} \in \mathbb{R}^2$ be expressed as a linear combination of $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 4 \end{bmatrix}$?

13. (a) Let $\vec{u}, \vec{w} \in \mathbb{R}^n$ with $\vec{w} \neq \vec{0}$ be such that $\vec{u}$ and $\vec{w}$ are not orthogonal. Prove that

$$\text{proj}_{(\text{proj}_{\vec{w}}\ \vec{u})}\ \vec{u} = \text{proj}_{\vec{w}}\ \vec{u}.$$

(b) Let $A \in M_{m\times n}(\mathbb{R})$ and $B \in M_{n\times m}(\mathbb{R})$. Define

$$C = A^T A + BB^T.$$

Prove that C is symmetric, that is, prove that $C^T = C$. Clearly justify your work.